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A small molecule targeting hepatitis B surface antigen inhibits clinically relevant drug-resistant hepatitis B virus

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Background: Currently approved oral antivirals for chronic HBV infection target the reverse transcriptase (RT) domain of the HBV polymerase. Emergence of drug resistance has been reported in a small proportion of chronic HBV patients on prolonged treatment with antivirals. We recently reported ZINC20451377, a small molecule targeting hepatitis B surface antigen (HBsAg) that effectively inhibits both WT HBV and tenofovir-resistant HBV. Due to the partial overlap between the RT domain and HBsAg, drug-resistant mutants are associated with corresponding mutations in HBsAg.

Objectives: To evaluate the efficacy of ZINC20451377 against nine clinically relevant drug-resistant HBV mutants that lead to simultaneous mutations in the overlapping HBsAg gene.

Methods: Huh7 cells were transfected with 1.2× HBV replicons corresponding to WT HBV or drug-resistant HBV mutants and treated with different concentrations of ZINC20451377. We assessed the IC_{50} values of ZINC20451377 for HBsAg levels in the culture supernatants using ELISAs. HBV secretion was measured by immunocapture of secreted virions followed by real-time PCR quantitation of virion-associated DNA.

Results: ZINC20451377 led to a dose-dependent inhibition of secreted HBsAg encoded by WT HBV and all nine drug-resistant mutants tested and the IC_{50} values were in the low micromolar range. ZINC20451377 inhibited HBV secretion from drug-resistant mutants except for mutants harbouring the rtL180M+rtM204V (MV) mutation.

Conclusions: The small molecule ZINC20451377 inhibits HBsAg and virion secretion in some of the clinically relevant drug-resistant HBV mutants. ZINC20451377 has a modest overall effect, and it was not effective against the MV mutants (lamivudine- and entecavir-resistant mutants).

Introduction

Chronic hepatitis B (CHB) remains a global problem despite the availability of an effective vaccine. CHB increases the risk of liver cancer by more than 30-fold[1] and is one of the leading causes of liver cirrhosis. Treatment of CHB with antivirals effectively suppresses HBV DNA levels, but long-term treatment may lead to the emergence of drug-resistant mutants. Tenofovir and entecavir are the most widely used antivirals for treating CHB because of their high efficacy and a relatively high genetic barrier for resistance compared with other antivirals.[2] About 1% of nucleoside-naive patients were found to develop resistance after 5 years of entecavir treatment,[3] but resistance to tenofovir disoproxil fumarate is rare.[4]

Hepatitis B surface antigen (HBsAg) is the envelope protein of HBV and the earliest detectable serological marker of HBV infection.[5] HBsAg is among the most abundantly produced HBV proteins, and it elicits a neutralizing antibody response. In HBV infection, HBsAg is secreted as subviral particles in vast excess compared with the infectious virion. Subviral particles can inhibit HBV-specific immune response and may serve as decoys for neutralizing antibodies, making them important players in establishing chronic HBV infection.[6] The presence of HBsAg in serum is an independent risk factor for development of hepatocellular carcinoma (HCC).[7] Mutations in HBsAg are associated with immune escape and occult HBV infection and may alter the clinical outcome of CHB.[8] Despite sustained viral suppression by the antivirals, a 'functional cure' defined as HBsAg clearance is rarely achieved. Therefore, HBsAg inhibitors may have potential therapeutic applications.

The emergence of drug-resistant mutations in HBV is often associated with clinical complications.[9] Hence, it is important to evaluate new molecules for activity against drug-resistant HBV. We have recently identified a small molecule (ZINC20451377, a piperazine derivative) targeting HBsAg using computational screening, docking and molecular dynamics simulations. ZINC20451377 binds to HBsAg and effectively inhibits the tenofovir-resistant CYEI mutant HBV at low micromolar concentration.[10] We also demonstrated that this small molecule is not cytotoxic and neither does it interfere with cellular transcription or translation.[10] One or more substitutions in the HBV reverse transcriptase (RT) have been associated with resistance to anti-HBV drugs. As the surface ORF overlaps with the HBV polymerase's RT region, mutations within the RT region resulting in drug resistance may also lead to amino acid substitutions in HBsAg. Well-documented HBV drug-resistant mutants, rtS106C, rtD134E, rtV173L and rtM204V (mutations in the HBV RT), lead to sL98V, sI126S, sE164D and sI195M substitutions respectively in HBsAg or the surface (s) protein.[11] Some of these HBsAg mutations impact HBV replication and have important clinical implications. For example, the sI126S and the sL98V mutations can impair virion secretion and alter antigenicity of HBsAg.[12,13] Treatment-induced mutations in HBsAg, sE164D and sI195M, are immune-escape mutants associated with reduced binding to anti-HBs antibody.[14] It is therefore particularly important to assess the efficacy of anti-HBV agents targeting HBsAg on drug-resistant mutants. Here, we evaluate the efficacy of the small molecule ZINC20451377 on nine clinically relevant drug-resistant HBV mutants using 1.2× HBV replicons.

Materials and methods

Molecules

ZINC20451377, a small molecule targeting HBsAg,[10] was procured from ChemBridge, USA (http://www.chembridge.com) and tenofovir disoproxil fumarate from Gilead Sciences, USA.

Plasmid constructs

The 1.2× HBV replicons (genotype C)[11] carrying nine different combinations of the RT mutations (rtS106C, rtH126Y, rtD134E, rtV173L, rtL180M and rtM204V) were kind gifts from Professor Kyun-Hwan Kim (Konkuk University, South Korea). All mutations were confirmed by sequencing. Additional details on the 1.2× HBV replicons used in this study are given in Table S1, available as Supplementary data at *JAC* Online.

Transfection of HBV constructs into Huh7 cells

Huh7 cells were seeded overnight in a 48-well plate at a density of 5×10^4 cells/well and transfected with the 1.2× HBV replicons using Lipofectamine 2000. Six hours post-transfection, media containing different concentrations of ZINC20451377 dissolved in DMSO (0.5% final concentration) were added; culture supernatants were harvested after 48 h. Transfections were performed in triplicate.

Estimation of secreted HBsAg

Secreted HBsAg levels were estimated using Bio-Rad MONOLISA HBsAg Ultra ELISA kits. Huh7 cells transfected with the 1.2× HBV replicon without the molecule (with a final concentration of 0.5% DMSO) were used as controls to assess the efficacy of the small molecule ZINC20451377.

Dose–response plots were generated using GraphPad Prism software version 8.1.1. The IC_{50} values were calculated using non-linear regression curve fitting and the 'log(inhibitor) versus response-variable slope (four parameters)' equation. IC_{90} values were calculated based on IC_{50} using GraphPad QuickCalcs (https://www.graphpad.com/quickcalcs/Ecanything1/). ELISAs were performed in triplicate and data are represented as mean ± SD.[15]

Western blot

Huh7 cells were transfected with WT 1.2× HBV replicon and treated with 10 μM ZINC20451377 for 48 h in a 6-well plate. For Western blot analysis, a goat polyclonal antibody to HBsAg was used as the primary antibody (Cat# BS-1557G, Thermo Fisher), and HRP-conjugated donkey anti-goat antibody was used as secondary antibody (Cat# 705-035-003, Jackson ImmunoResearch Laboratories). The level of HBsAg was normalized against alpha-tubulin.

Quantification of secreted hepatitis B virions

Huh7 cells were transfected with 1.2× HBV replicons and treated with 10 μM ZINC20451377 or tenofovir for 72 h in a 6-well plate. Quantification of secreted HBV was done using a technique developed in house.[16] In this method, virions are immunocaptured, followed by extraction of virion-associated DNA and real-time PCR. Primers specifically designed to amplify the virion-associated DNA were used as described previously.[17] A standard curve was generated for absolute quantification.[16,18]

Results and discussion

The small molecule ZINC20451377 binds HBsAg and inhibits its production and secretion.[10] ZINC20451377 clearly inhibits all three forms of HBV surface (small, middle and large) proteins (Figure S1). We evaluated the efficacy of the small molecule on nine drug-resistant 1.2× HBV replicons (see Table S1 for details) in Huh7 cells. The 1.2× replicons used in this study have been previously characterized for resistance to anti-HBV drugs.[11] To validate the HBV replicons used in our experimental settings, we used tenofovir. As expected, tenofovir significantly inhibited virion secretion from the WT and all drug-resistant mutants other than CYE, CYEMV and CYELMV (intermediate tenofovir resistance has been documented for these three mutants) (Figure S2).

The small molecule ZINC20451377 led to a dose-dependent inhibition of secreted HBsAg in all nine drug-resistant mutants tested (Figure 1). At 10 μM concentration, ZINC20451377 led to modest inhibition of HBsAg secretion in some drug-resistant mutants (CE, MV, CYEMV), resulting in higher IC_{50} values (Figure 1). IC_{90} values for inhibition of secreted HBsAg are listed in Table S2.

The small molecule ZINC20451377 was able to inhibit hepatitis B virion secretion from the WT, C, E, CE, YE, CY and CYE mutants (Figure 2). ZINC20451377 did not lead to a significant reduction in virion secretion from HBV mutants MV, CYEMV and CYELMV, in spite of inhibiting HBsAg from these mutants (albeit with higher IC_{50} values). Interestingly, ZINC20451377 could inhibit hepatitis B virion secretion from all drug-resistant HBV tested except for mutants harbouring the MV (rtL180M+rtM204V) mutation.

The rtM204V mutation in the YMDD active site of the RT domain of the HBV polymerase is a primary drug-resistance mutation that confers lamivudine resistance, but this mutation is associated with poor replication capacity.[19] rtL180M is a compensatory mutation that occurs concomitantly with rtM204V in the MV mutant, restoring the replication capacity of the

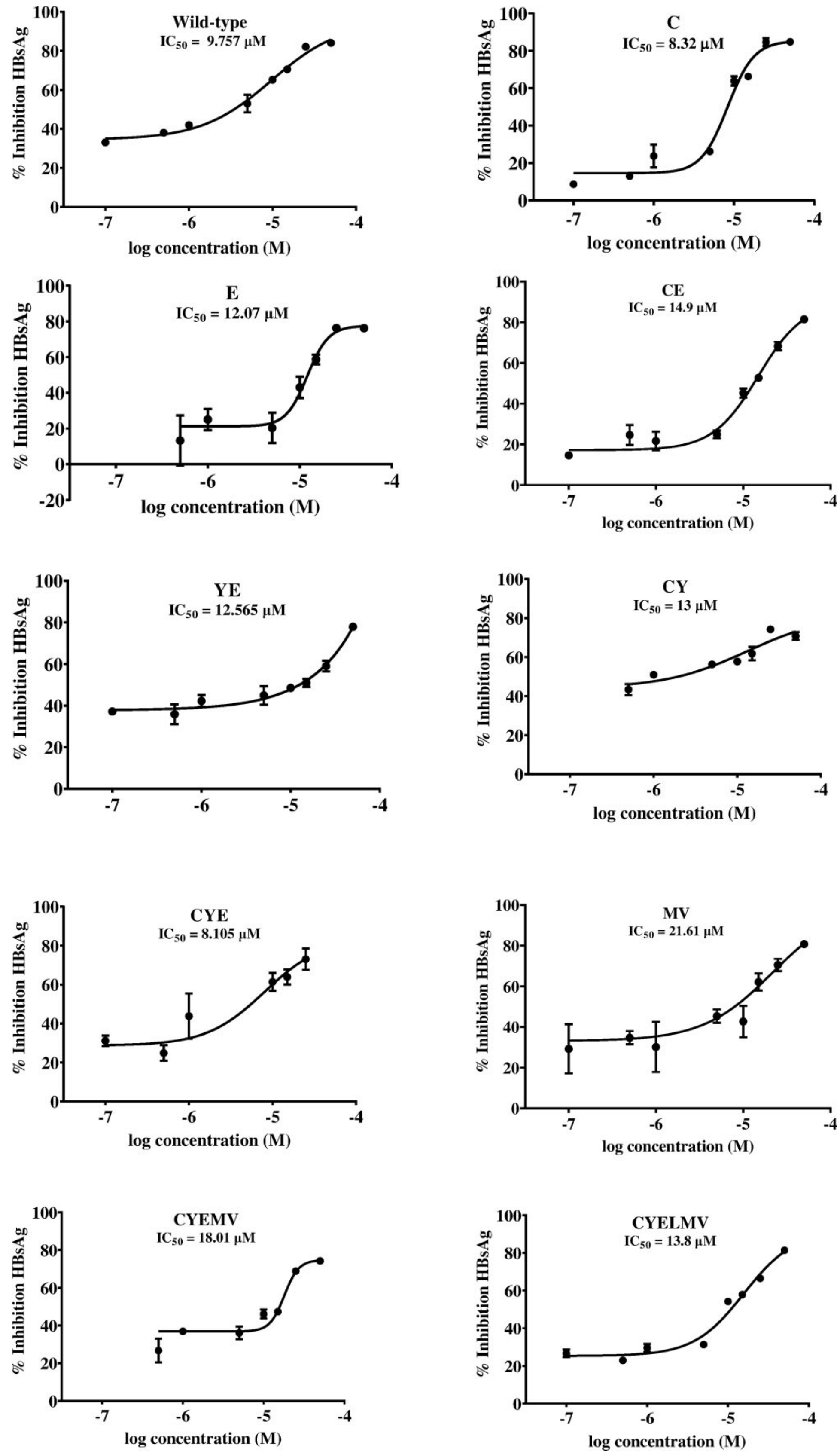

Figure 1. ZINC20451377 inhibits secreted HBsAg encoded by WT and drug-resistant HBV replicons in a dose-dependent manner from Huh7 cells. Details of the drug-resistant mutants are provided in Table S1.

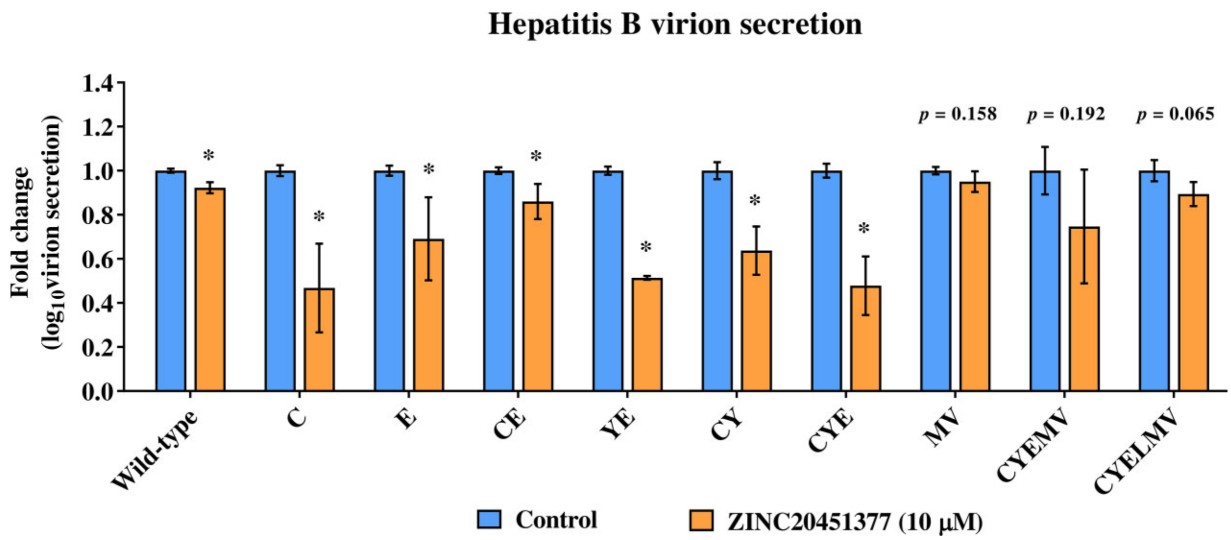

Figure 2. Fold change in secreted hepatitis B virions ($\log_{10}$) per mL supernatant from WT, C, E, CE, YE, CY, CYE, MV, CYEMV and CYELMV mutant HBV replicons after treatment with ZINC20451377 at 10 μM concentration as compared with DMSO-treated (vehicle) controls. *P* values <0.05 were considered significant (Student's *t*-test). Details of the drug-resistant mutants are provided in Table S1. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

rtM204V mutant to about 40% of that of the WT HBV. The addition of another compensatory mutation, rtV173L, restores the replication capacity of the MV mutant to 90% of that of WT HBV.[20] The LMV (rtV173L + rtL180M + rtM204V) mutant is resistant to lamivudine as well as entecavir and corresponds to an overlapping double mutation (sE164D and sI195M) in the HBV surface protein. Of note, ZINC20451377 could inhibit HBsAg from the HBV replicons having the MV mutant (MV, CYEMV) or the LMV mutant (CYELMV, CYELMVI) despite mutations in HBsAg due to the overlapping ORFs in HBV. While the compensatory nature of rtL180M and rtV173L and their ability to enhance HBV replication is well documented, the underlying mechanisms are not fully understood. We speculate that the capacity of the MV and the LMV mutants to enhance replication competence and the yet unknown underlying mechanisms may explain, at least in part, the inability of the small molecule ZINC20451377 to inhibit virion secretion from these mutants.

High levels of HBsAg in the serum often indicate poor disease prognosis as HBsAg can interfere with HBV-specific immune response[6] and also promote HCC development.[7] Although rarely achieved, the clearance of serum HBsAg is an important therapeutic goal in CHB treatment. HBsAg inhibitors capable of reducing the HBsAg levels have clinical significance as they may facilitate host-mediated clearance of serum HBsAg and therefore improve clinical outcomes in CHB patients.

In summary, the small molecule ZINC20451377 inhibits secreted HBsAg encoded by nine clinically relevant drug-resistant HBV mutants tested in this study at low micromolar concentrations, despite the presence of overlapping mutations in the HBV surface protein. In addition, the small molecule significantly inhibits virion secretion from some clinically relevant drug-resistant HBV mutants. The molecule has a modest overall effect and does not affect virion secretion in mutants harbouring the MV mutation that confers lamivudine and entecavir resistance. ZINC20451377 exhibits promising anti-HBV activity against some drug-resistant mutants and merits further investigation.

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Transparency declarations

None to declare.

Author contributions

Conceptualization: S.K, R.B., B.J. and P.V.; investigations: S.K.; writing—original draft preparation: S.K.; writing—review and editing, B.J. and P.V.; supervision: B.J. and P.V. All authors have read and agreed to the final version of the manuscript.

Supplementary data

Tables S1 and S2 and Figures S1 and S2 are available as Supplementary data at *JAC* Online.

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